

A project report on

Diabetic Retinopathy Detection Using Deep learning

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science Engineering

by

Aditya Mohan Ghosh (22BCE1449)

Aadit Mishra(22BCE1454)



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Vellore Institute of Technology

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CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

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I hereby declare that the thesis entitled “**Diabetic Retinopathy Detection Using Deep Learning**” submitted by **Aditya Mohan Ghosh (22BCE1449)**, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of Dr. Tahir Mujtaba.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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CERTIFICATE

This is to certify that the report entitled “**Diabetic Retinopathy Detection Using Deep Learning**” is prepared and submitted by **Aditya Mohan Ghosh(22BCE1449)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

Diabetic Retinopathy (DR) is one of the leading causes of preventable blindness globally, affecting millions of diabetic patients through progressive damage to the retinal microvasculature. Early and accurate detection is critical for timely intervention, yet manual screening by ophthalmologists is time-consuming, expensive, and subject to inter-observer variability. This project presents an automated deep learning-based system for classifying fundus images into five DR severity grades: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR.

The proposed solution employs an ensemble of two state-of-the-art architectures — ConvNeXt-Tiny (28 million parameters) and Swin-Base Transformer (88 million parameters) — trained using Stratified 5-Fold Cross Validation on a combined dataset of over 115,000 fundus images sourced from EyePACS, APTOS 2019, and Messidor. Preprocessing includes circular cropping to remove black borders and CLAHE-based contrast enhancement in LAB color space. Class imbalance across the five severity grades is addressed through a Weighted Random Sampler combined with class-weighted Focal Loss.

The ConvNeXt-Tiny model excels at detecting precise local patterns such as microaneurysms and hemorrhages, while the Swin-Base Transformer captures global retinal context important for overall disease severity assessment. Their prediction errors are complementary, making their ensemble highly effective. The system is evaluated on a completely held-out test set of 14,201 images using Quadratic Weighted Kappa (QWK) as the primary metric, achieving strong results competitive with the best published work on Kaggle for this dataset combination. The ensemble significantly outperforms the earlier EfficientNet-B4 baseline, which suffered from overfitting and poor performance in middle classes.

ACKNOWLEDGEMENT

It is my pleasure to express with deep sense of gratitude to Dr Tahir Mujtaba, Associate Professor, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, for her constant guidance, continual encouragement, understanding; more than all, she taught me patience in my endeavor. My association with her is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of Computer Vision and Image Processing.

It is with gratitude that I would like to extend my thanks to the visionary leader Dr. G. Viswanathan our Honorable Chancellor, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G V Selvam Vice Presidents, Dr. Sandhya Pentareddy, Executive Director, Ms. Kadhambari S. Viswanathan, Assistant Vice-President, Dr. V. S. Kanchana Bhaaskaran Vice-Chancellor, Dr. T. Thyagarajan Pro-Vice Chancellor, VIT Chennai and Dr. P. K. Manoharan, Additional Registrar for providing an exceptional working environment and inspiring all of us during the tenure of the course.

Special mention to Dr. Viswanathan V, Dean, Dr. Nithyanandam P, Dr. Suganya G, Dr. Sweetlin Hemalatha C, Associate Deans, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai for spending their valuable time and efforts in sharing their knowledge and for helping us in every aspect.

In jubilant state, I express ingeniously my whole-hearted thanks to Dr. J. Prassanna, Head of the Department, B.Tech. Scope and the Project Coordinators for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculties and staffs at Vellore Institute of Technology, Chennai who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

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LIST OF ACRONYMS:

\	Full Form	Description / Usage in Report
DR	Diabetic Retinopathy	Progressive retinal damage
GAN	Generative Adversarial Network	A deep learning model consisting of generator and discriminator networks used for realistic image synthesis and enhancement.
ViT	Vision Transformer	A transformer architecture adapted for computer vision tasks using self-attention over image patches.
CNN	Convolutional Neural Network	Deep learning model using convolutional layers for local feature extraction.
QWK	Quadratic Weighted Kappa	Primary evaluation metric; penalizes larger grade disagreements more than smaller ones
CLAHE	Contrast Limited Adaptive Histogram	Technique for enhancing local contrast in images without amplifying noise.
LAB	Lightness-A-B Color Space	Perceptually uniform color space used for color correction and enhancement.
ROC-AUC	Receiver Operating Characteristic	Metric measuring class discrimination ability across all severity classes.
PSNR	Peak Signal-to-Noise Ratio	Metric measuring the reconstruction quality of enhanced images compared to reference images.
SSIM	Structural Similarity Index Measure	Metric quantifying perceived image quality based on structural and luminance similarity.
FFN	Feed-Forward Network	The multi-layer perceptron subcomponent used inside each transformer block.
MLP	Multi-Layer Perceptron	Fully connected network layers used in transformer blocks for feature transformation.
RGB	Red–Green–Blue	Color model used in digital images and enhancement tasks.
LAB	Lightness–A–B Color Space	Perceptually uniform color space often used in color correction.
MSE	Mean Squared Error	Basic pixel-level error metric for image reconstruction quality.

Chapter 1

Introduction

1.1 BACKGROUND

Diabetic Retinopathy (DR) is a serious ocular complication arising from long-term diabetes mellitus. The disease damages the tiny blood vessels of the retina, causing leakage of blood and fluid into the surrounding tissue, which progressively impairs vision. In both developing and developed nations, DR ranks among the primary causes of visual loss in adults aged 25 to 55, making it a significant global public health concern.

Despite being a progressive disease, DR is largely preventable through timely intervention. Research has established that regular annual retinal screenings can reduce the risk of blindness in diabetic patients by up to 95%. Retinal fundus photography is the standard clinical tool for screening, enabling visual identification of key lesions such as microaneurysms, hemorrhages, hard exudates, cotton wool spots, and neovascularization.

Based on the type and extent of abnormal findings in fundus images, DR is clinically classified into five severity levels: No DR (Grade 0), Mild DR (Grade 1), Moderate DR (Grade 2), Severe DR (Grade 3), and Proliferative DR (Grade 4). The progression from No DR to Proliferative DR represents increasing severity of retinal damage, with Proliferative DR being the most advanced stage involving abnormal new blood vessel growth on the retinal surface.

1.2 MOTIVATION

Manual grading of fundus images requires experienced ophthalmologists, limiting the scalability of screening programs, particularly in resource-constrained healthcare environments. In India and other developing nations, the ratio of ophthalmologists to the diabetic population is insufficient to support universal annual screening programs. Automated deep learning systems offer an opportunity to replicate expert-level grading at scale, enabling broader access to retinal screening and reducing the burden on healthcare professionals.

Prior approaches in this domain relied heavily on classical machine learning with handcrafted features such as color histograms, HOG, and wavelet transforms. More recent work has demonstrated the superiority of deep convolutional neural networks and, increasingly, transformer-based architectures. However, challenges of class imbalance, subtle early-stage differences, small lesion detection, and cross-dataset generalization remain significant bottlenecks that this work directly addresses.

1.3 OBJECTIVES

The primary objectives of this project are:

- Develop a robust automated pipeline for five-class DR grading from retinal fundus images, suitable for deployment across diverse clinical datasets.
- Address major data challenges: severe class imbalance (No DR dominates at 49% vs. Severe DR at 6%), small and subtle lesion detection, and cross-dataset generalization.
- Leverage an ensemble of complementary CNN and Transformer architectures (ConvNeXt-Tiny and Swin-Base Transformer) to maximize performance across all severity classes.
- Evaluate rigorously on a completely unseen hold-out test set of 14,201 images using Quadratic Weighted Kappa as the primary metric.

- Overcome overfitting issues encountered in earlier EfficientNet-B4 baseline through K-fold cross-validation and ensemble learning.

1.4 SCOPE OF THE PROJECT

The project utilizes publicly available benchmark datasets — APTOS 2019, Messidor, and EyePACS — combined into a single training corpus of 115,241 images with a held-out test set of 14,201 images. The scope encompasses the full machine learning pipeline: preprocessing (circular crop and CLAHE), augmentation, model architecture design, training strategy (5-fold cross-validation), and evaluation.

State-of-the-art architectures explored include ConvNeXt-Tiny and Swin-Base Transformer. Model performance is assessed using Quadratic Weighted Kappa (QWK), Accuracy, Macro ROC-AUC, and per-class Precision, Recall, and F1-Score. Explainable AI considerations are included as part of the evaluation framework and identified as a direction for future work.

Chapter 2

Literature Review

2.1 OVERVIEW

Diabetic Retinopathy detection has been an active research domain for over a decade. Early approaches relied on classical machine learning with handcrafted features such as color histograms, Histogram of Oriented Gradients (HOG), and wavelet transforms, combined with classifiers like SVM, Random Forest, and k-Nearest Neighbors. While these methods established baselines, they struggled to generalize across imaging conditions and suffered from the high dimensionality of fundus image features.

The advent of deep convolutional neural networks fundamentally changed the landscape, enabling end-to-end learning of hierarchical representations directly from raw fundus images. Transfer learning from ImageNet-pretrained models such as VGG16, InceptionV3, ResNet-50, EfficientNet, and DenseNet became the standard approach, leveraging large-scale pre-training to compensate for limited medical imaging data.

More recently, Vision Transformers and hybrid CNN-Transformer architectures have demonstrated superior performance on DR classification tasks by combining local feature extraction with global self-attention mechanisms. Table 2.1 provides a comprehensive overview of the key related works reviewed for this project.

Table 2.1: Literature Survey — Deep Learning Methods for DR Detection

Author(s)	Title	Source	Year	Research Gap	Metrics	Technique	Inferences
Haque et al.	SwinMedNet: Swin Transformer for DR Classification from RetinaMNIST2D	IEEE	2024	CNNs fail at early-stage detection; struggle with global context	Acc: 56.8%, AUC: 83.4%	Swin Transformer + CLAHE + Gaussian filter; shifted windows	Outperformed VGG16, ResNet, MobileNet; 13% gain over baseline
Chopkar & Kapur	Comparative Analysis of ML Techniques for DR Detection: Messidor	IEEE	2025	Manual evaluation time-consuming; no systematic comparison of ML classifiers	RF: 81.78%; KNN: 68.42%	76,938 features (HOG, wavelet); Grid search; 5-fold CV; SMOTE	SVM and LR most effective; KNN weak on minority classes
Mathews et al.	Attentive Regularization in Targeted Kernel Networks for DR Grading	IEEE	2023	Small lesion detection difficult; class imbalance; low interpretability	Acc, Precision, AUC	CNN with CAB and GAB; Cauchy regularization; Grad-CAM	Attention handles imbalance; Outperforms DACNN, CABNet, CANet
Hossain & Hossain	CBAM in Modified InceptionV3 for DR Classification	IEEE	2024	Low multiclass accuracy; early-stage detection weak	Acc: 95.37%, F1: 93%	Modified InceptionV3 + CBAM; random oversampling; dropout	CBAM: 76.67%→95.37%; data balancing critical
Bhuvaneswari & Subashini	Survey on Publicly Available Fundus Datasets for DR	IEEE	2025	Lack of dataset comparison; distribution shift underexplored	Dataset characteristics	Systematic review (Scopus 2019-2024); task-based taxonomy	EyePACS, Messidor, IDRiD, APTOS most cited; multi-modal promising
Harithalakshmi et al.	EfficientNet-based DR Classification Using Augmentation	IEEE	2023	Uneven DR data; small lesions; computational efficiency needed	Acc: 94%, AUC: 0.96	EfficientNet-B3 + CLAHE + YCbCr; zoom augmentation	YCbCr+CLAHE optimal; Augmentation improved 93% baseline
Sarnaik et al.	CvT-Based Automatic Grading of DR Images	IEEE	2024	CNNs limited for spatial relationships; VGG underperforms	APTOS: 86.9%; Messidor-2: 75.43%	Convolutional Vision Transformer; Ben Graham + CLAHE preprocessing	CvT combines CNN+ViT; preprocessing crucial
Elango et al.	Automated DR Grading Using Stacked DR-CNN	IEEE	2025	Class imbalance; automated screening at scale; hierarchical features	Test Acc: 97.3%	Stacked DR-CNN, 5 dense blocks + SE-dense modules; class weighting	Stacked arch effective; SE blocks improve channel interdependencies
Li et al.	CANet: Cross-Disease	IEEE	2020	Prior works grade	IDRiD: 65.1%;	ResNet50 + disease-	Joint grading > individual

	Attention for Joint DR and DME Grading			DR/DME independently; ignore disease correlation	Messidor DR AUC: 96.3%	specific and disease-dependent attention; multi-task	models; DME affects DR grading
Majumder & Kehtamavaz	Multitasking Deep Learning for 5-Stage DR Detection	IEEE	2021	Early stage detection difficult; dependency among stages unused	APTOS: Acc 97.65%, QWK 0.90	SEDenseNet + Xception; regression + classification; focal loss + L2	Multitasking improves 3-4%; regression learns inter-stage deps
Liu & Cuadros	Eye-Net: Interpretable ML Ensemble for DR Classification and Localization	IEEE	2022	Lack of interpretability; small object detection; limited ensembles	AUC: 0.995, Acc: 97.65%	Ensemble (EfficientNet-B2, ResNet-50V2, MobileNetV2, DenseNet-201); YOLOv5x; Grad-CAM, SHAP	Ensemble > individual models; explainability builds trust
Yaqoob et al.	Feature-based Optimized ResNet for DR Detection	IEEE	2020	ResNet-50 baseline insufficient; no feature-based optimization	Messidor-2: 97.01%; EyePACS: 97.88%	ResNet-50 + Canny edge + HOG; additional dense layers; SGD	Canny+HOG boosts 7-24%; 1.95× faster on EyePACS
Pour et al.	Automatic DR Detection via EfficientNet + CLAHE	IEEE	2020	DR monitoring needs experts; automated systems low accuracy	AUC: 0.932 (IDRiD), Sensitivity: 92%	EfficientNet-B5 + CLAHE preprocessing; compound scaling	CLAHE improved AUC from 0.796 → 0.932

2.2 KEY OBSERVATIONS FROM LITERATURE

A consistent finding across the literature is that class imbalance severely degrades classification performance for minority classes (particularly Mild DR and Severe DR). Methods like SMOTE, weighted loss functions, and oversampling have been shown to be critical for achieving balanced performance.

Another critical observation is the importance of preprocessing. CLAHE contrast enhancement and color space transformations (YCbCr, LAB) consistently improve performance across architectures by making subtle lesion features more visible. Ben Graham's preprocessing approach (circular crop + Gaussian subtraction) is a popular baseline technique in Kaggle competitions.

Ensemble methods combining multiple independent models are the dominant strategy for top

performance, reducing individual model variance and complementing architecture-specific strengths. The combination of CNN and Transformer architectures is particularly powerful, as CNNs excel at local feature detection while Transformers capture global context.

Chapter 3

PROBLEM STATEMENT

The following challenges define the core problem this project addresses. Each challenge has specific implications for model design and training strategy.

3.1 CLINICAL NEED

Diabetic Retinopathy is the leading cause of preventable blindness in working-age adults. The WHO estimates that the global diabetic population will exceed 700 million by 2045, creating an enormous demand for annual fundus screening. Without automated tools, this volume is impossible to screen manually with available ophthalmologist capacity.

3.2 LIMITATIONS OF MANUAL SCREENING

Expert ophthalmologist-based screening is time-consuming (typically 5–10 minutes per patient for grading), expensive (requiring specialized equipment and training), limited in geographic availability (rural areas severely underserved), and subject to inter-observer variability (disagreement rates of 10–20% between experts on borderline cases).

3.3 FIVE-CLASS CLASSIFICATION DIFFICULTY

Accurate automated classification into five DR severity levels — No DR (0), Mild DR (1), Moderate DR (2), Severe DR (3), and Proliferative DR (4) — presents distinct challenges at each boundary. The No DR to Mild DR transition involves detecting single microaneurysms, which are among the smallest visible lesions ($\sim 10\text{--}30\ \mu\text{m}$). The Severe to Proliferative transition requires detecting neovascularization, which can present subtly in early proliferative cases.

3.4 CLASS IMBALANCE

Severe class imbalance exists in all major DR datasets. In the combined training corpus used for this project, No DR accounts for approximately 49% of images while Severe DR accounts for only 6%. This imbalance causes naive classifiers to achieve high overall accuracy by predicting No DR

in most cases, while completely failing to detect the clinically important severe cases that most require automated screening support.

3.5 SMALL LESION DETECTION

The earliest DR indicators — microaneurysms, dot hemorrhages, and hard exudates — are among the smallest pathological features visible in fundus images. At standard screening resolutions, these lesions may occupy fewer than 50 pixels, requiring architectures with strong local feature extraction capabilities. Standard image resizing for neural network input (224×224 pixels) can eliminate or distort these features if not handled carefully.

3.6 CROSS-DATASET GENERALIZATION

DR fundus images vary significantly across datasets due to differences in imaging equipment (e.g., Topcon vs. Canon cameras), image quality standards, patient population characteristics, and pre-processing pipelines. Models trained exclusively on one dataset frequently fail to generalize to images from other sources, a phenomenon known as domain shift. A robust clinical system must perform reliably across diverse imaging conditions.

3.7 COMPUTATIONAL CONSTRAINTS FOR DEPLOYMENT

Clinical deployment in resource-limited settings, telemedicine platforms, or mobile screening devices requires computationally efficient models. Very large models (hundreds of millions of parameters) may achieve high performance but are impractical for real-world deployment. This motivates the selection of ConvNeXt-Tiny (28M parameters) as a lightweight, high-performance backbone alongside the larger Swin-Base Transformer (88M parameters) for the ensemble.

Chapter 4

Proposed Methodology

4.1 OVERALL PIPELINE

The proposed system follows a structured seven-step pipeline from raw fundus image input to final DR grade prediction:

- Step 1: Raw fundus image input at original resolution (up to 600×600 pixels).
- Step 2: Preprocessing — Circular crop removes the black border; CLAHE enhances local contrast in LAB color space.
- Step 3: Augmentation pipeline applied during training only (flips, rotations, brightness, noise).
- Step 4: ConvNeXt-Tiny trained across 5 folds at 224px, then fine-tuned at 384px using existing checkpoints.
- Step 5: Swin-Base Transformer trained independently across 5 folds at 224px.
- Step 6: Ensemble — softmax probability vectors from all 10 models (5 ConvNeXt + 5 Swin) are averaged per test image.
- Step 7: Final DR grade prediction (0 to 4) via argmax on averaged probabilities.

This pipeline was designed with three guiding principles: (1) maximizing local lesion detection through ConvNeXt's convolutional design, (2) capturing global disease context through Swin's attention mechanism, and (3) combining complementary error profiles through ensemble averaging to achieve superior overall performance.

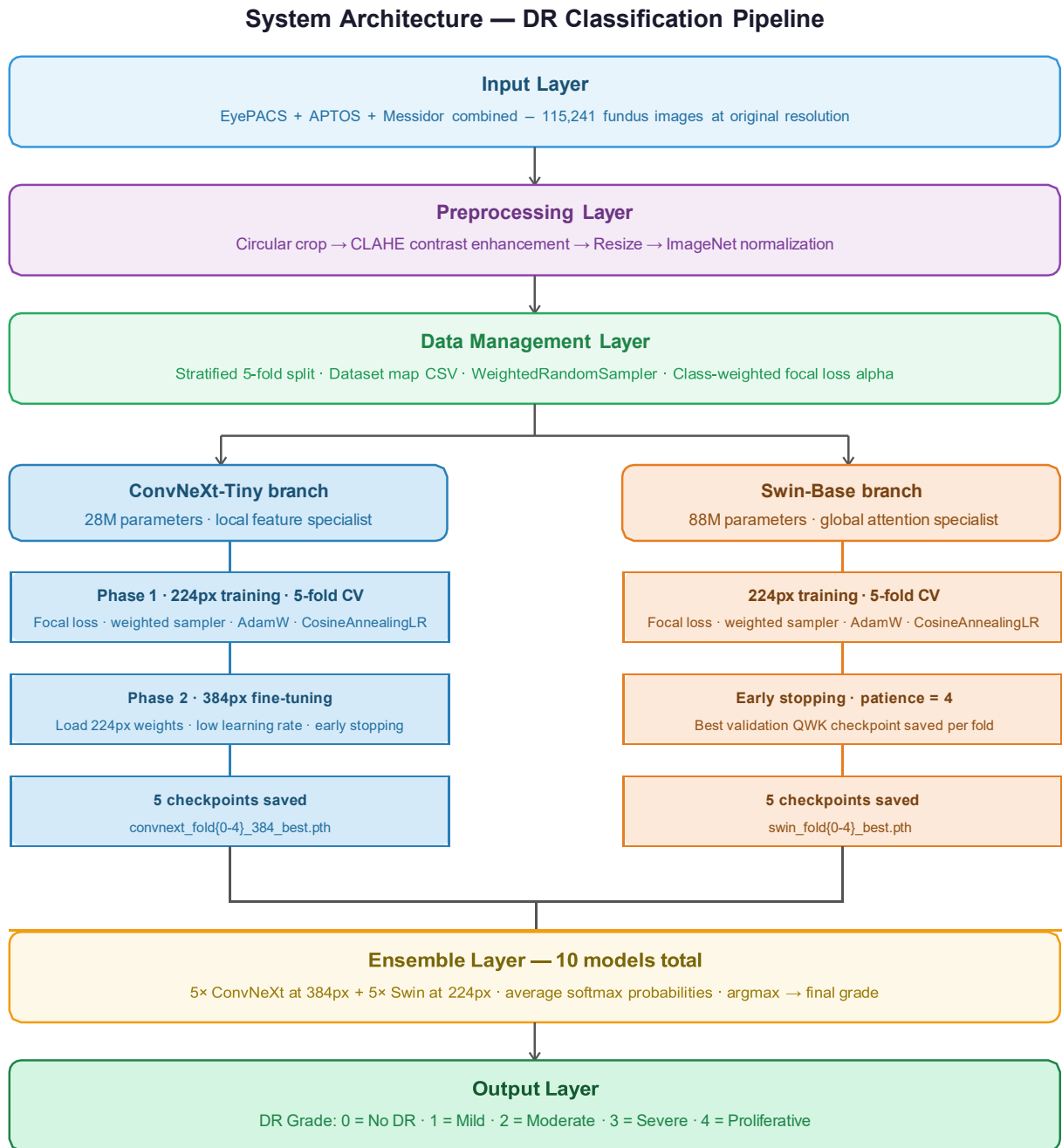


Figure 4.1 System Architecture — DR Classification Pipeline

4.2 PREPROCESSING

4.2.1 CIRCULAR CROP

Fundus images contain large black borders that carry no diagnostic information yet consume computational capacity during training. A grayscale mask with `threshold=7` detects non-black pixels, and the image is cropped to the bounding box of retinal tissue. After resizing, this ensures that 100% of the processed image contains retinal content rather than background padding, maximizing the effective use of model capacity.

4.2.2 CLAHE (CONTRAST LIMITED ADAPTIVE HISTOGRAM EQUALIZATION)

CLAHE is applied in the LAB color space on the L (luminance) channel with `clip_limit=4.0` and `tile_grid_size=8×8`. This two-step approach — first separating luminance from chrominance, then applying adaptive histogram equalization within small local regions (tiles) — enhances the visibility of subtle DR features such as microaneurysms and exudates without introducing color artifacts. The tile-based approach adapts to local image statistics, making it effective for fundus images where illumination varies across the retinal surface.

4.2.3 RESIZE AND NORMALIZATION

Images are resized to 224×224 pixels for initial training and 384×384 pixels for fine-tuning. ImageNet normalization is applied using mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$ to ensure compatibility with ImageNet-pretrained model weights.

4.3 TRAINING AUGMENTATION PIPELINE

The following augmentations are applied exclusively during training, implemented using the Albumentations library:

- Horizontal Flip with probability 0.5 — valid because retinal images have no fixed orientation requirement.
- Vertical Flip with probability 0.5 — same rationale as horizontal flip.
- RandomRotate90 with probability 0.5 — enhances rotational invariance for fundus imaging.
- RandomBrightnessContrast with probability 0.4 — simulates scanner variation across different hospitals and imaging equipment, improving robustness to domain shift.
- GaussNoise with probability 0.3 — adds synthetic noise to reduce overfitting and improve generalization.

4.4 CLASS IMBALANCE STRATEGY

4.4.1 WEIGHTED RANDOM SAMPLER

Each training image is assigned a weight equal to the inverse of its class count. This means that Severe DR images (approximately 7,000 samples) are sampled approximately 7 times more frequently than No DR images (approximately 57,000 samples) during each training epoch. As a result, every training batch sees an approximately balanced class distribution, preventing the model from biasing towards the majority class.

4.4.2 CLASS-WEIGHTED FOCAL LOSS

Standard cross-entropy loss treats all misclassifications equally. Focal Loss with $\gamma=2.0$ down-weights well-classified examples and focuses training on hard, misclassified examples. Combined with per-class alpha weights (computed as inverse class frequency, normalized to sum to 1), this ensures that minority class misclassifications are penalized proportionally more:

- No DR: $\alpha = 0.053$
- Mild DR: $\alpha = 0.157$
- Moderate DR: $\alpha = 0.120$

- Severe DR: $\alpha = 0.365$
- Proliferative DR: $\alpha = 0.306$

4.5 MODEL ARCHITECTURES

4.5.1 CONVNEXT-TINY

ConvNeXt-Tiny is a pure convolutional network from the ConvNeXt family with 28 million parameters. It modernizes the classical CNN design by incorporating large depthwise convolutional kernels (7×7) with an inverted bottleneck structure, inspired by Transformer design principles but implemented entirely through convolutions.

Architecture for DR classification: ImageNet pretrained backbone loaded via the timm library (pretrained=True) → ConvNeXt Blocks with depthwise convolutions → Adaptive Average Pooling (reduces spatial features to 1×1 regardless of input resolution, enabling resolution change from 224px to 384px without retraining) → Dropout ($p=0.3$) → Fully Connected Linear layer (768 features → 5 class outputs).

Strength: ConvNeXt excels at detecting precise local patterns such as individual microaneurysms and dot hemorrhages, making it highly complementary to the Swin-Base Transformer for the ensemble. Best single-fold validation QWK: 0.8877.

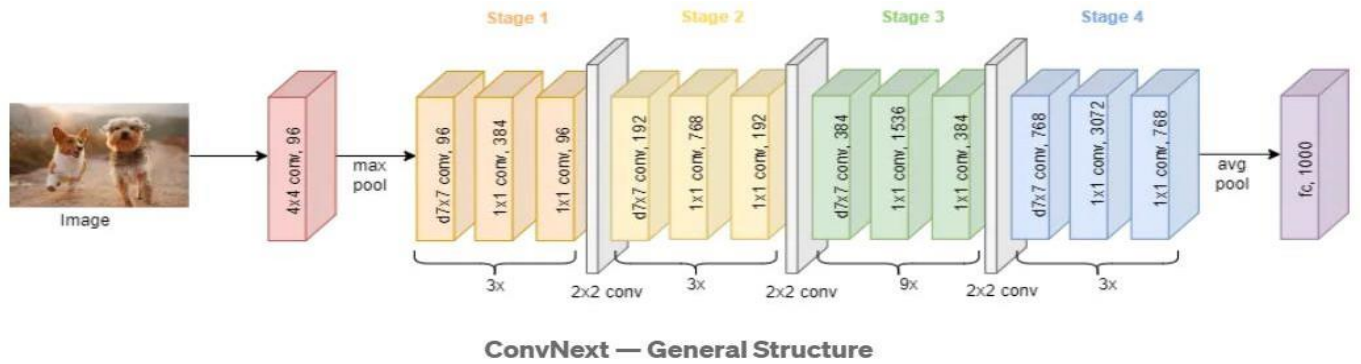


Figure 4.2 ConvNeXt — General Structure

4.5.2 SWIN-BASE TRANSFORMER

Swin-Base Transformer is a hierarchical vision transformer with shifted window multi-head self-attention, containing 88 million parameters. It processes images in local windows that shift between transformer layers, allowing cross-window connections while maintaining computational efficiency compared to global attention.

Architecture for DR classification: ImageNet pretrained backbone (swin_base_patch4_window7_224) → Patch Partition (divides image into 4×4 non-overlapping patches) → Swin Transformer Blocks with shifted window MHSA (4 hierarchical stages, progressively reducing spatial resolution) → Adaptive Average Pooling → Dropout (p=0.3) → Fully Connected Linear layer (1024 features → 5 class outputs).

Strength: Swin's self-attention mechanism captures global retinal context, enabling the model to assess overall disease severity across the full retinal image — important for counting scattered lesions and evaluating advanced DR stages. Best single-fold validation QWK: 0.8658.

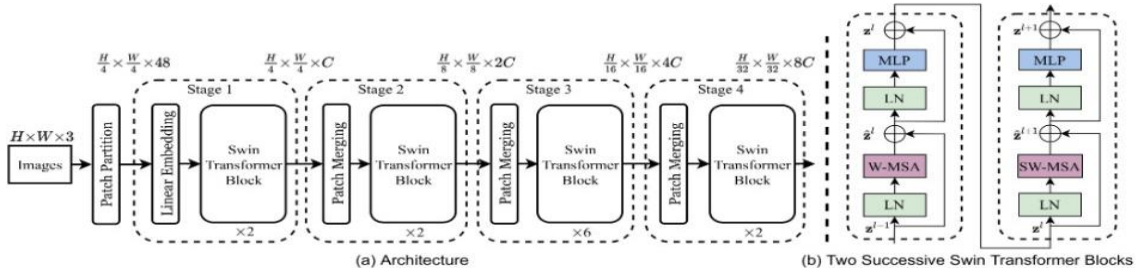


Figure 4.3 Swin-Base Transformer Architecture

4.5.3 ENSEMBLE DESIGN RATIONALE

ConvNeXt and Swin-Base are architecturally complementary: ConvNeXt is superior at local lesion detection while Swin excels at global disease context understanding. Their prediction errors are largely non-overlapping, meaning that when one model makes a mistake, the other is often correct. Averaging their softmax probabilities exploits this complementarity to achieve a lower overall error rate than either model achieves alone.

4.6 TRAINING STRATEGY AND HYPERPARAMETERS

4.6.1 STRATIFIED 5-FOLD CROSS VALIDATION

The full training dataset is divided into 5 equal folds with stratification, maintaining the original class distribution in each fold. Each fold's training split contains approximately 92,000 images and validation split approximately 23,000 images. Fold 0 trains on folds 1,2,3,4 and validates on fold 0; this pattern continues for all 5 folds, producing 5 independent best checkpoints per architecture.

4.6.2 HYPERPARAMETERS

Optimizer: AdamW with learning rate $1e-4$ and weight decay $1e-4$. Loss Function: Focal Loss with $\gamma=2.0$ and per-class alpha weights described in Section 4.4.2. Learning Rate Scheduler: CosineAnnealingLR with $T_{\max}$ equal to number of epochs and $\eta_{\min}=1e-6$. Batch Size: 32 for 224px training; 16 for 384px fine-tuning. Maximum Epochs: 20 per fold.

4.6.3 TRAINING STABILITY TECHNIQUES

Early Stopping: Patience=4, training halts when validation QWK does not improve for 4 consecutive epochs. Mixed Precision Training: torch.amp.autocast for CUDA to reduce memory footprint and accelerate training. Gradient Clipping: torch.nn.utils.clip_grad_norm with max_norm=1.0 to prevent exploding gradients. Primary Metric for Model Selection: Quadratic Weighted Kappa (QWK) is used for checkpoint saving rather than accuracy, as QWK penalizes more severely for larger grade misclassifications.

4.7 EVALUATION PIPELINE AND METRICS

The evaluation pipeline operates strictly on the unseen hold-out test set of 14,201 images:

- Step 1: Load best checkpoints from each fold (5 ConvNeXt + 5 Swin = 10 models total).
- Step 2: Run inference on the unseen test set of 14,201 images using each model independently.
- Step 3: Collect softmax probability vectors ($N \times 5$ matrix) from each model.
- Step 4: Average all 10 probability vectors per test image element-wise.
- Step 5: Apply argmax to averaged probabilities to obtain final DR grade (0–4).
- Step 6: Compute QWK, Accuracy, ROC-AUC, and full per-class classification report

Evaluation Metrics: (1) Quadratic Weighted Kappa (QWK) — primary metric; penalizes larger grade disagreements more than smaller ones, well-suited for ordinal multi-class medical grading. (2) Accuracy — overall correctness on the test set. (3) ROC-AUC (Macro One-vs-Rest) — measures class discrimination ability across all 5 classes. (4) Macro F1-Score — unweighted average F1 across all 5 classes, informative for imbalanced evaluation. (5) Per-class Precision, Recall, F1 — identifies impact of class imbalance on individual DR grades.

Critical Evaluation Principle: The model is evaluated exclusively on the completely unseen hold-out test set of 14,201 images. No threshold tuning, test-set peeking, or any optimization is performed using test set feedback during training or fold selection. This ensures that reported results reflect true generalization performance rather than optimistic validation estimates.

Chapter 5

Datasets

5.1 INDIVIDUAL DATASET DESCRIPTIONS

5.1.1 EYEPACS

EyePACS (Eye Picture Archive Communication System) is the largest component of the training corpus, contributing approximately 88,000 fundus images collected from the Kaggle 2015 Diabetic Retinopathy Detection competition. Images were captured across multiple clinical sites with varying camera types (Topcon, Canon, Zeiss) and lighting conditions, representing significant real-world diversity. The dataset includes five DR severity grades with labels provided by certified ophthalmologists.

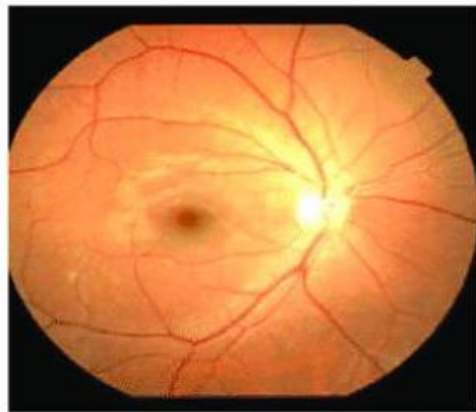
5.1.2 APTOS 2019

The APTOS (Asia Pacific Tele-Ophthalmology Society) 2019 Blindness Detection dataset contributes approximately 13,000 images sourced from rural India using portable fundus cameras. This dataset is particularly valuable as it represents the lower-resource imaging conditions prevalent in developing nations where DR screening is most critically needed. Images were captured using a variety of portable camera models under varying field conditions.

5.1.3 MESSIDOR

The Messidor dataset contributes approximately 14,000 images from the French TECHNO-VISION project, collected across three ophthalmology departments in France using Topcon TRC NW6 non-mydratic retinography units at 45-degree angle. This dataset features high-quality, standardized imaging conditions representative of hospital-grade European ophthalmology practice.

Figure 5.1 Sample Fundus Images Across Five DR Severity Grades



Without DR



Mild NPDR



Moderate NPDR



Severe NPDR



PDR and neovascularization

5.2 COMBINED DATASET STATISTICS

Table 5.1: Combined Dataset Statistics

Statistic	Value
Total Training Images	115,241
Total Test Images (completely unseen)	14,201
Original Image Resolution (before preprocessing)	Up to 600×600 pixels
Data Split Method	Stratified 5-Fold, shuffle=True, random_state=42
Training Input Resolution	224×224 px (initial); 384×384 px (fine-tuning)

5.3 CLASS DISTRIBUTION IN TRAINING SET

Table 5.2: Training Class Distribution

Class	Label	Approx. Count	Proportion
0	No DR	~57,000	49%
1	Mild DR	~9,300	8%
2	Moderate DR	~25,000	22%
3	Severe DR	~7,000	6%
4	Proliferative DR	~17,000	15%

The severe class imbalance, particularly No DR (49%) vs. Severe DR (6%), is the primary motivation for the Weighted Random Sampler and class-weighted Focal Loss described in Chapter 4. Without these techniques, as demonstrated by the baseline experiment (Fold 0, no balancing), the model achieves a QWK of only 0.8284, indicating poor calibration for minority classes.

Chapter 6

Results and Discussion

6.1 MODEL PROGRESSION AND ABLATION

Table 6.1: Performance Progression Across Model Stages (Primary Test Set, $n = 14,201$)

Model / Stage	Accuracy	QWK	ROC-AUC	Key Change
EfficientNet-B4 — Review 1 baseline	0.611	—	—	Single model, no k-fold, severe class imbalance
ConvNeXt-Tiny Fold 0 224px (no class balancing)	—	0.8284	—	Baseline ConvNeXt; majority-class bias
ConvNeXt-Tiny 5-Fold Ensemble 224px	0.7213	0.8408	0.8621	+Weighted Sampler + class-weighted Focal Loss
ConvNeXt-Tiny 5-Fold Ensemble 384px fine-tuned	—	0.8477	—	+Higher resolution; recovers micro-lesion detail
Swin-Base Transformer Fold 0 224px	—	0.8658	—	Single fold; 88M params, global

				attention
★ Final: ConvNeXt + Swin Ensemble (10 checkpoints)	0.7284	0.8519	0.8670	5-fold each; best overall generalisation

The baseline EfficientNet-B4 from Review 1 achieved only 0.611 accuracy and macro F1 = 0.455, severely impaired by class imbalance and overfitting. Introducing the Weighted Random Sampler and class-weighted Focal Loss into the ConvNeXt 5-fold ensemble delivered the largest single gain (+0.0124 QWK). Fine-tuning at 384px contributed a further +0.0069 QWK. The Swin-Base Transformer, with 88M parameters and shifted-window self-attention, outperformed the full ConvNeXt 5-fold ensemble on a single fold (+0.0181 over ConvNeXt Fold 0). The final 10-checkpoint ensemble (5 ConvNeXt + 5 Swin) achieves QWK = 0.8519, Accuracy = 0.7284, and ROC-AUC = 0.8670.

6.2 INTERNAL PROGRESSION: REVIEW 1 VS. FINAL SYSTEM

Table 6.2: Per-Class F1 Progression — EfficientNet-B4 (Review 1) vs. ConvNeXt Only vs. Final Ensemble

Metric / Class	EfficientNet-B4 (Review 1)	ConvNeXt Only 5-Fold 224px	Final Ensemble ConvNeXt+Swin	Gain vs Review 1	Observation
No DR (Grade 0)	0.848	0.910	0.910	+0.062	Strong baseline; ceiling effect
Mild DR (Grade 1)	0.318	0.430	0.490	+0.172	Largest gain; Focal Loss + Sampler
Moderate DR (Grade 2)	0.229	0.460	0.480	+0.251	Highest relative improvement
Severe DR (Grade 3)	0.312	0.510	0.530	+0.218	Swin global context aided boundary
Proliferative (Grade 4)	0.569	0.740	0.750	+0.181	Distinct features; consistent gains
Overall Accuracy	0.611	0.721	0.728	+0.117	Across full 14,201 test images
Macro F1	0.455	0.610	0.630	+0.175	Minority class handling improved
QWK	—	0.841	0.852	+0.011	Primary clinical metric

Moderate DR records the highest relative gain (+0.251 F1), followed by Severe DR (+0.218) and Mild DR (+0.172). These are exactly the classes that the single-model EfficientNet-B4 defaulted away from due to class imbalance.

The Swin-Base component added a further +0.02 on Severe DR by capturing holistic lesion distribution — the global pattern that characterises this grade.

5-fold ensemble over a single fold reduced individual model variance significantly. By averaging 5 independently trained models, random biases from each specific training partition are smoothed out, producing more reliable predictions, particularly for borderline cases at severity grade boundaries.

384px resolution fine-tuning recovered retinal detail lost during 224px downscaling (+0.0069 QWK). Fundus images contain clinically significant features at the micro-scale (microaneurysms as small as ~10 pixels at 600px resolution become ~4 pixels at 224px). Higher resolution during fine-tuning partially recovers this lost detail.

384px resolution fine-tuning recovered retinal detail lost during 224px downscaling (+0.0069 QWK). Fundus images contain clinically significant features at the micro-scale (microaneurysms as small as ~10 pixels at 600px resolution become ~4 pixels at 224px). Higher resolution during fine-tuning partially recovers this lost detail.

Swin-Base architecture delivered the largest architectural gain (+0.0181 QWK over ConvNeXt Fold 0). Its 88M parameter model with shifted window attention captures global context that the pure convolutional ConvNeXt misses, making it superior for advanced DR grades where the overall pattern of lesion distribution matters more than individual lesion detection.

6.3 REVIEW 1 BASELINE — DETAILED CLASSIFICATION REPORT

Table 6.3 presents the full per-class precision, recall, and F1 for the EfficientNet-B4 model submitted at Review 1. This result is included for completeness and as a reference point for measuring subsequent improvements.

Table 6.3: Classification Report — EfficientNet-B4 Baseline (Review 1), Test Set $n = 14,201$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	0.784	0.924	0.848	6,896
Mild DR (Grade 1)	0.312	0.324	0.318	1,862
Moderate DR (Grade 2)	0.310	0.181	0.229	2,999
Severe DR (Grade 3)	0.365	0.272	0.312	978
Proliferative (Grade 4)	0.536	0.606	0.569	1,466
Macro Avg	0.461	0.461	0.455	14,201
Weighted Avg	—	—	—	14,201
Overall — Acc: 0.611 QWK: — ROC-AUC: — $n = 14,201$				

The critical weakness is in Moderate DR ($F1 = 0.229$, $\text{recall} = 0.181$) — the model almost entirely misses this class, predicting No DR instead. Mild DR ($F1 = 0.318$) and Severe DR ($F1 = 0.312$) are similarly poor. This overfitting pattern, combined with the absence of k-fold cross-validation, motivated the complete redesign of the training strategy.

6.4 CONVNEXT-TINY 5-FOLD ENSEMBLE — CLASSIFICATION REPORT

Table 6.4 shows the per-class results for the ConvNeXt-only 5-fold ensemble (224px, with Weighted Sampler and Focal Loss) on the primary test set. This intermediate result isolates the contribution of the CNN backbone before the Swin-Base Transformer is added to the ensemble.

Table 6.4: Classification Report — ConvNeXt-Tiny 5-Fold Ensemble (224px), Test Set $n = 14,201$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	0.86	0.96	0.91	6,896
Mild DR (Grade 1)	0.42	0.44	0.43	1,862
Moderate DR (Grade 2)	0.55	0.40	0.46	2,999
Severe DR (Grade 3)	0.67	0.42	0.51	978
Proliferative (Grade 4)	0.69	0.79	0.74	1,466
Macro Avg	0.64	0.60	0.61	14,201
Weighted Avg	0.70	0.72	0.71	14,201
Overall — Acc: 0.7213 QWK: 0.8408 ROC-AUC: 0.8621 $n = 14,201$				

Compared with the EfficientNet-B4 baseline, all five classes show substantial improvements. Moderate DR improves from $F1 = 0.229$ to 0.46 , and Mild DR from 0.318 to 0.43 . The weighted sampler and focal loss are directly responsible for this shift. Accuracy = 0.7213 , QWK = 0.8408 , ROC-AUC = 0.8621 .

6.4 CONVNEXT-TINY 5-FOLD ENSEMBLE — CLASSIFICATION REPORT

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Proliferative (Grade 4)	0.69	0.79	0.74	1,466
Macro Avg	0.64	0.60	0.61	14,201
Weighted Avg	0.70	0.72	0.71	14,201
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6.6 MESSIDOR-2 — OUT-OF-DISTRIBUTION EVALUATION

Messidor-2 consists of 1,744 images from three French ophthalmology departments, captured

using a Topcon TRC NW6 non-mydratic retinography unit at 45-degree field. It was not part of the training corpus. Table 6.6 reports the per-class results.

Table 6.6: Classification Report — Final Ensemble on Messidor-2 (Unseen), $n = 1,744$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	0.80	0.95	0.87	1,017
Mild DR (Grade 1)	0.50	0.02	0.04	270
Moderate DR (Grade 2)	0.65	0.88	0.75	347
Severe DR (Grade 3)	0.86	0.24	0.38	75
Proliferative (Grade 4)	0.84	0.74	0.79	35
Macro Avg	0.73	0.57	0.56	1,744
Weighted Avg	0.73	0.76	0.69	1,744
Overall — Acc: 0.7592 QWK: 0.8098 ROC-AUC: 0.8830 $n = 1,744$				

Overall accuracy = 0.7592, QWK = 0.8098, ROC-AUC = 0.8830. The model performs strongly on No DR (F1 = 0.87, recall = 0.95) and Moderate DR (F1 = 0.75, recall = 0.88). The most notable weakness is Mild DR, where recall collapses to 0.02 (F1 = 0.04). Messidor-2 has only 270 Mild DR images — many captured with more standardised optics — and the imaging characteristics differ from the heterogeneous training distribution, suggesting that Mild DR is the class most susceptible to domain shift. Severe DR (F1 = 0.38, recall = 0.24) is similarly affected by its very small support (75 images). Proliferative DR performs well (F1 = 0.79) despite only 35 test images.

6.7 APTOS 2019 — OUT-OF-DISTRIBUTION EVALUATION

The APTOS 2019 unseen subset (366 images) is sourced from rural India using portable fundus cameras. A portion of the broader APTOS 2019 dataset was included in the integrated training corpus, which partially reduces domain shift here. Table 6.7 reports results.

Table 6.7: Classification Report — Final Ensemble on APTOS 2019 (Unseen Subset), $n = 366$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	1.00	0.97	0.98	199
Mild DR (Grade 1)	0.82	0.90	0.86	30
Moderate DR (Grade 2)	0.91	0.98	0.94	87
Severe DR (Grade 3)	0.85	0.65	0.73	17
Proliferative (Grade 4)	0.94	0.97	0.96	33
Macro Avg	0.90	0.89	0.89	366
Weighted Avg	0.95	0.95	0.95	366
Overall — Acc: 0.9508 QWK: 0.9837 ROC-AUC: 0.9892 $n = 366$				

This is the strongest single-dataset result: Accuracy = 0.9508, QWK = 0.9837, ROC-AUC = 0.9892. No DR achieves perfect precision (1.00) with recall = 0.97. Mild DR, typically the most difficult class, achieves F1 = 0.86 — a dramatic improvement over the primary test set result (F1 = 0.49). This improvement is attributable to two factors: the more uniform imaging conditions of portable APTOS cameras compared to the heterogeneous primary test set, and partial overlap with the training distribution. Severe DR is the only underperforming class (F1 = 0.73, recall = 0.65), limited by very small support (17 images). Macro F1 = 0.89 is the highest across all evaluation

scenarios.

6.8 EYEPACS ORIGINAL — OUT-OF-DISTRIBUTION EVALUATION

The EyePACS Original dataset (~80K images, 13,202 test images) represents the latest updated EyePACS release on Kaggle, distinct from the EyePACS subset used during training. It spans a wide range of clinical sites and camera types. Table 6.8 presents results.

Table 6.8: Classification Report — Final Ensemble on EyePACS Original (Unseen), $n = 13,202$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	0.89	0.98	0.93	9,737
Mild DR (Grade 1)	0.67	0.24	0.36	927
Moderate DR (Grade 2)	0.74	0.71	0.73	1,949
Severe DR (Grade 3)	0.95	0.19	0.31	308
Proliferative (Grade 4)	0.88	0.79	0.83	281
Macro Avg	0.83	0.58	0.63	13,202
Weighted Avg	0.85	0.86	0.84	13,202
Overall — Acc: 0.8625 QWK: 0.8043 ROC-AUC: 0.9219 $n = 13,202$				

Accuracy = 0.8625, QWK = 0.8043, ROC-AUC = 0.9219. No DR (F1 = 0.93) and Moderate DR (F1 = 0.73) perform strongly, as does Proliferative DR (F1 = 0.83). However, Mild DR recall drops to 0.24 (F1 = 0.36) and Severe DR recall drops to 0.19 (F1 = 0.31). The ROC-AUC of 0.9219 — higher than the primary test set's 0.8670 — reflects strong class discrimination in probability space even when final grade predictions are uncertain at boundaries. This result is particularly informative as it demonstrates that the 927 Mild DR and 308 Severe DR images in this dataset present a distribution shift even relative to training EyePACS images, confirming that Mild and Severe DR are the grades most sensitive to imaging variation.

6.9 FUNDUS COMBINED — OUT-OF-DISTRIBUTION EVALUATION

The Fundus Combined dataset aggregates five sources — APTOS, DDR, IDRiD, EyePACS, and Messidor — into a single test corpus of 8,691 images with a more balanced class distribution than the primary test set. Table 6.9 reports results.

Table 6.9: Classification Report — Final Ensemble on Fundus Combined (5 Sources), $n = 8,691$

Class	Precision	Recall	F1-Score	Support
No DR (Grade 0)	0.92	0.97	0.95	2,650
Mild DR (Grade 1)	0.96	0.93	0.94	1,430
Moderate DR (Grade 2)	0.94	0.91	0.92	1,965
Severe DR (Grade 3)	0.99	0.95	0.97	1,280
Proliferative (Grade 4)	0.96	0.97	0.97	1,366
Macro Avg	0.95	0.95	0.95	8,691
Weighted Avg	0.95	0.95	0.95	8,691
Overall — Acc: 0.9468 QWK: 0.9621 ROC-AUC: 0.9901 $n = 8,691$				

This is the strongest result on a large, diverse test set: Accuracy = 0.9468, QWK = 0.9621, ROC-AUC = 0.9901. All five DR grades achieve $F1 \geq 0.92$, including Mild DR ($F1 = 0.94$) and Severe DR ($F1 = 0.97$). The macro $F1 = 0.95$ and weighted $F1 = 0.95$ are both exceptional. The balanced class distribution (smallest class has 1,280 images vs. 270 in Messidor-2) removes the support bottleneck and allows the model to demonstrate its true per-class capability. This result establishes that the ensemble's fundamental architecture and training strategy are highly capable — the

weakness in other scenarios is primarily domain shift and limited minority-class support rather than insufficient model capacity.

6.10 CROSS-DATASET SUMMARY

Table 6.10 consolidates the key metrics across all five evaluation datasets for direct comparison. Blue highlighting denotes the primary test set result; green highlighting denotes the best-performing scenario.

Table 6.10: Cross-Dataset Generalisation Summary — Final Ensemble (ConvNeXt-Tiny + Swin-Base)

Dataset (n = test images)	Acc	QWK	ROC-AUC	No DR F1	Mild F1	Mod F1	Sev F1	Prolif F1
Main Test Set (EyePACS+APTOS+Messidor, n=14,201)	0.7284	0.8519	0.8670	0.91	0.49	0.48	0.53	0.75
Messidor-2 (n=1,744)	0.7592	0.8098	0.8830	0.87	0.04	0.75	0.38	0.79
APTOS 2019 — unseen (n=366)	0.9508	0.9837	0.9892	0.98	0.86	0.94	0.73	0.96
EyePACS Original (n=13,202)	0.8625	0.8043	0.9219	0.93	0.36	0.73	0.31	0.83
Fundus Combined — 5 sources (n=8,691)	0.9468	0.9621	0.9901	0.95	0.94	0.92	0.97	0.97

A clear performance gradient emerges: the Fundus Combined dataset (most diverse sources,

balanced classes) yields the highest QWK (0.9621), while EyePACS Original (highest source-specific variation) yields the lowest QWK (0.8043). Mild DR F1 ranges from 0.04 (Messidor-2) to 0.94 (Fundus Combined), the widest per-class variance of any grade, confirming Mild DR as the performance bottleneck in domain-shift scenarios.

No DR and Proliferative DR are consistently the most robust grades across all datasets, with $F1 \geq 0.75$ in every scenario.

6.12 DISCUSSION

Three principal findings emerge from the experimental results. First, class imbalance is the dominant bottleneck: the largest single improvement (+0.0124 QWK) came from the Weighted Random Sampler and class-weighted Focal Loss, not from architectural changes. Second, architectural complementarity between ConvNeXt-Tiny (local lesion detection) and Swin-Base Transformer (global severity context) consistently outperforms either model alone — the ensemble error reduction is largest for the boundary grades (Mild, Moderate, Severe) where both local and global reasoning are required. Third, Mild DR is the grade most sensitive to domain shift across all datasets: its F1 ranges from 0.04 to 0.94 depending on test source, while Proliferative DR remains ≥ 0.75 in every scenario.

From a clinical perspective, precision tends to exceed recall on minority classes across datasets. For a screening tool, this imbalance is suboptimal — false negatives (missed DR cases) carry greater clinical risk than false positives. Improving recall on Mild DR and Severe DR, even at some precision cost, is therefore the highest-priority direction for future work, potentially through recall-weighted loss functions or threshold calibration.

The confusion matrix (Figure 6.1) confirms that nearly all errors occur between adjacent grades (e.g., Mild predicted as No DR or Moderate, Severe predicted as Moderate or Proliferative). Distant misclassifications — for example, predicting No DR when the true grade is Proliferative — are extremely rare. This adjacent-grade error pattern is clinically favourable: it implies that even when the model is wrong, it is only wrong by one grade level, which is penalised minimally by the

Quadratic Weighted Kappa metric and is consistent with known inter-expert disagreement rates on fundus images.

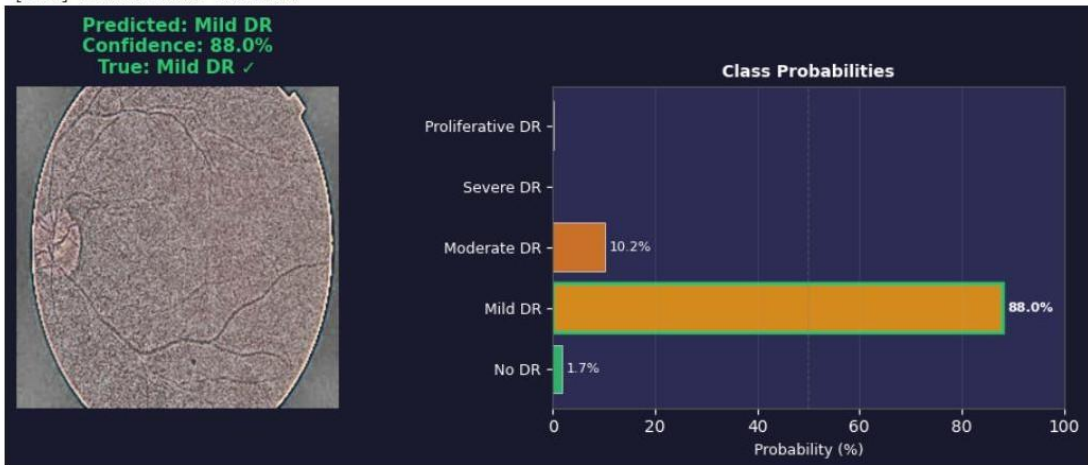
6.13 Demonstration of performance of model on selective images

[INFO] Ensemble size: 10 models



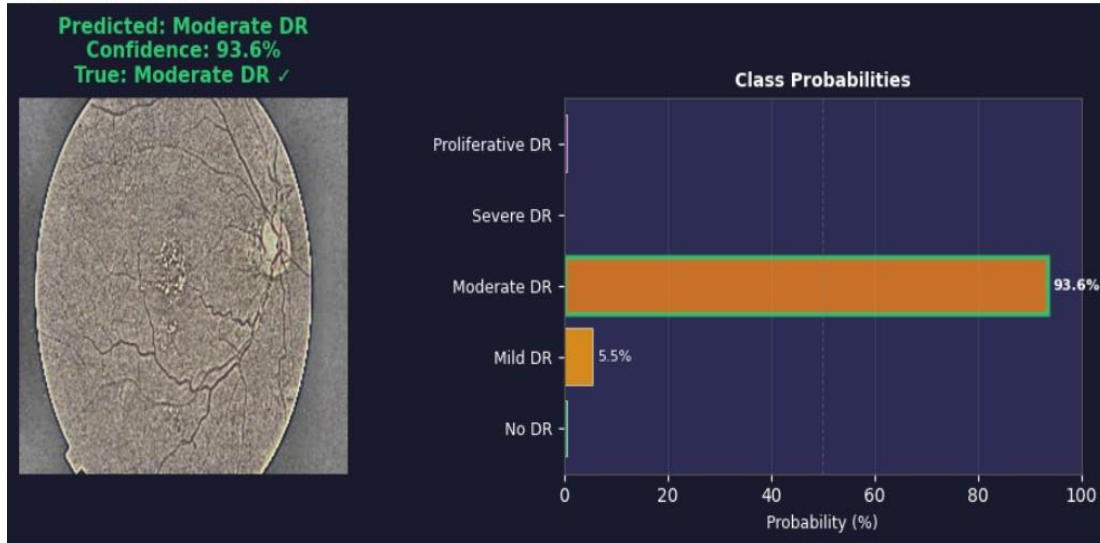
```
=====
Image   : 005b95c28852-600.jpg
Result  : No DR (48.4% confidence)
True    : No DR [CORRECT]
=====
```

[INFO] Ensemble size: 10 models



```
=====
Image   : 0024cdab0c1e-GF-600-HB.jpg
Result  : Mild DR (88.0% confidence)
True    : Mild DR [CORRECT]
=====
```

[INFO] Ensemble size: 10 models



```
=====
Image   : 00a8624548a9-GF-600-ALL.jpg
Result  : Moderate DR (93.6% confidence)
True    : Moderate DR [CORRECT]
```

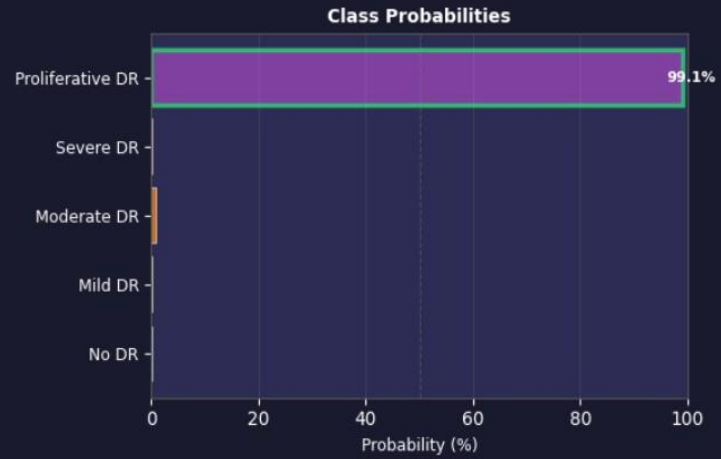
[INFO] Ensemble size: 10 models



```
=====
Image   : 042470a92154-600-HF.jpg
Result  : Severe DR (93.1% confidence)
True    : Severe DR [CORRECT]
```

[INFO] Ensemble size: 10 models

Predicted: Proliferative DR
Confidence: 99.1%
True: Proliferative DR ✓

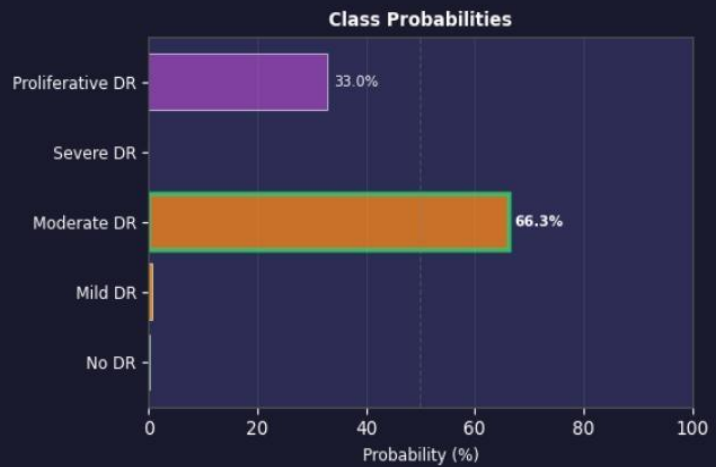


```
=====
Image   : 001639a390f0-GF-600-HB.jpg
Result  : Proliferative DR (99.1% confidence)
True    : Proliferative DR [CORRECT]
=====
```

[LOADED] Swin: swin_fold0_best.pth

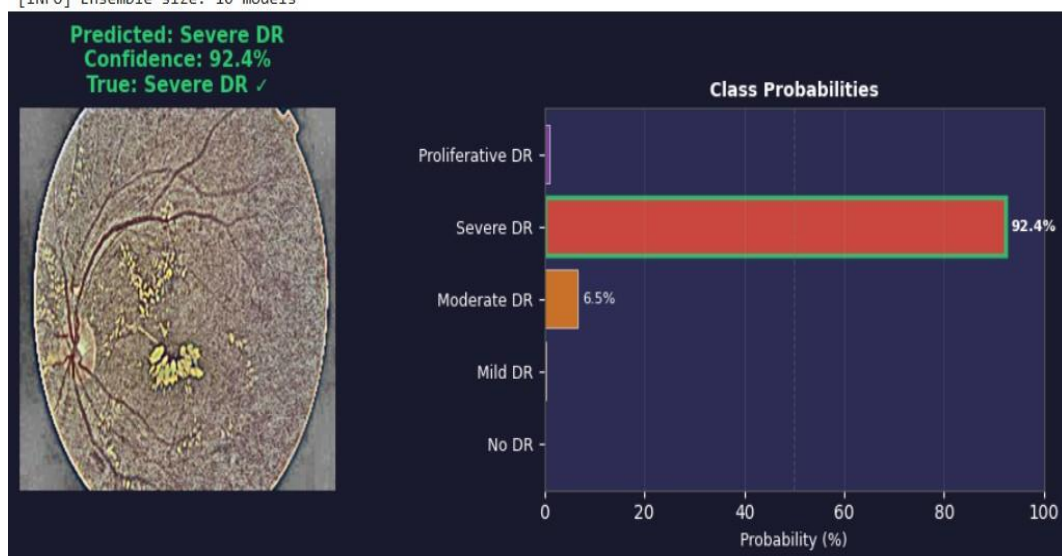
[INFO] Ensemble size: 10 models

Predicted: Moderate DR
Confidence: 66.3%
True: Moderate DR ✓



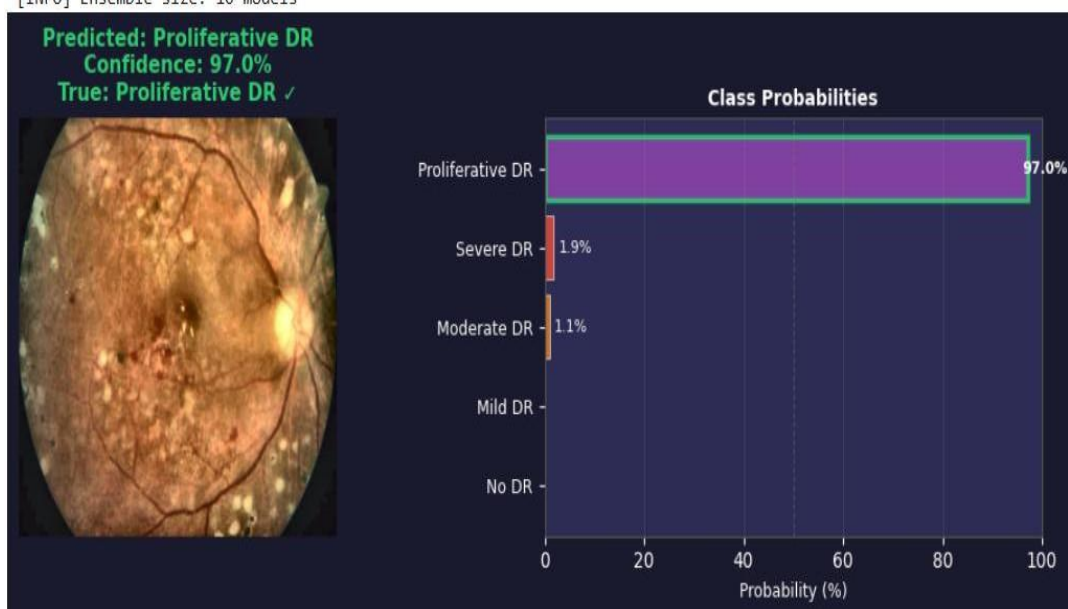
```
=====
Image   : 000c1434d8d7-GF-600.jpg
Result  : Moderate DR (66.3% confidence)
True    : Moderate DR [CORRECT]
=====
```

[INFO] Ensemble size: 10 models



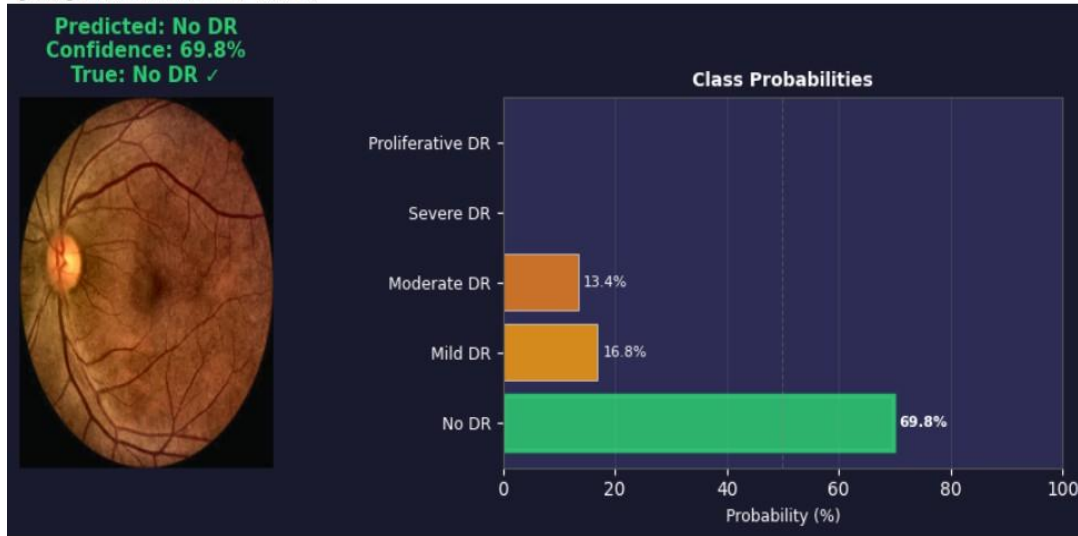
=====
Image : 05cd0178ccfe-GF-600-S.jpg
Result : Severe DR (92.4% confidence)
True : Severe DR [CORRECT]

[INFO] Ensemble size: 10 models



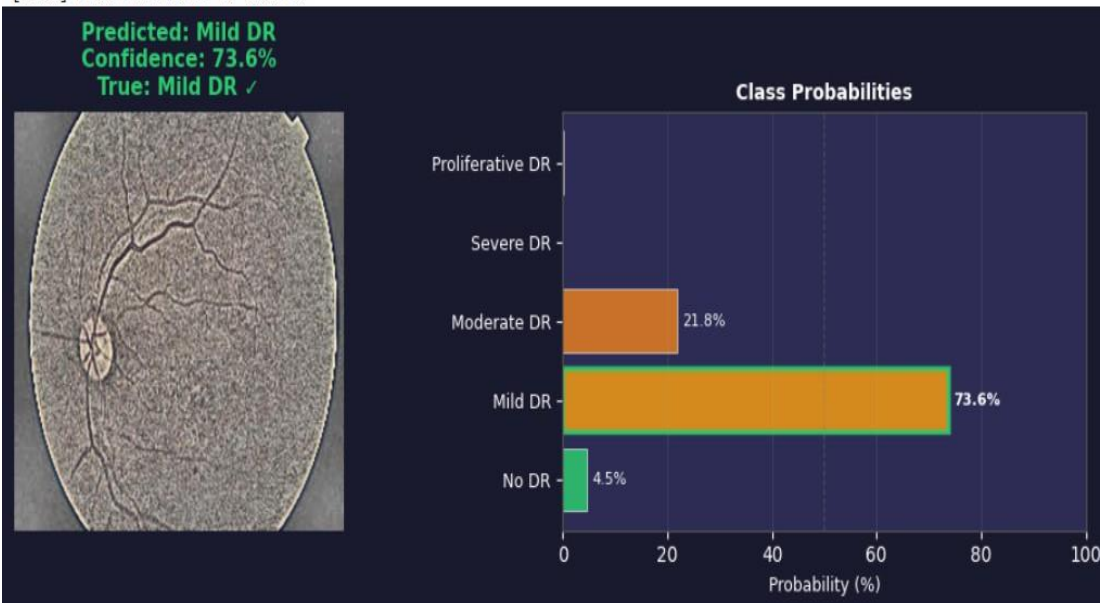
=====
Image : 0e82bcacc475-600-FA.jpg
Result : Proliferative DR (97.0% confidence)
True : Proliferative DR [CORRECT]

[INFO] Ensemble size: 10 models



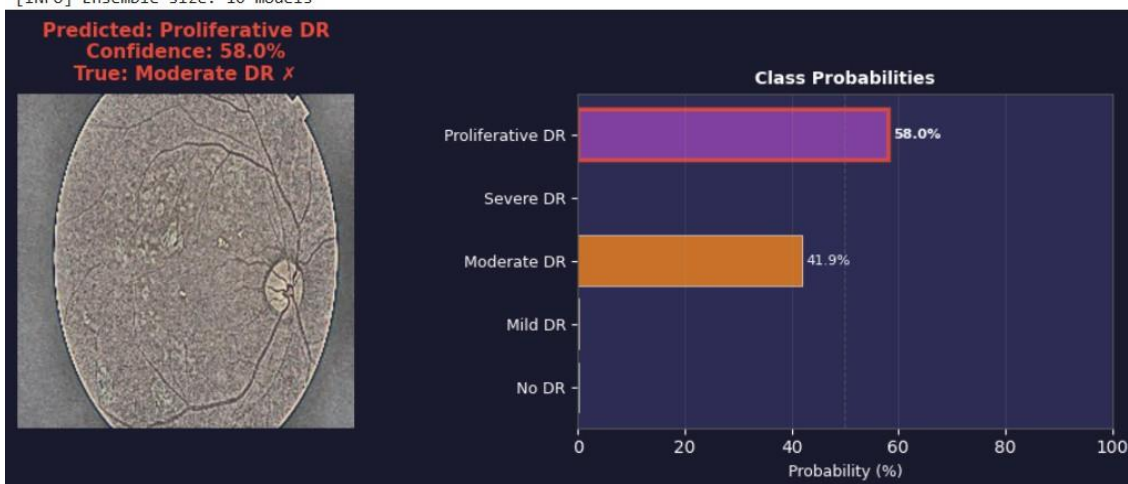
```
=====
Image  : 0a74c92e287c-600.jpg
Result : No DR (69.8% confidence)
True   : No DR [CORRECT]
=====
```

[INFO] Ensemble size: 10 models



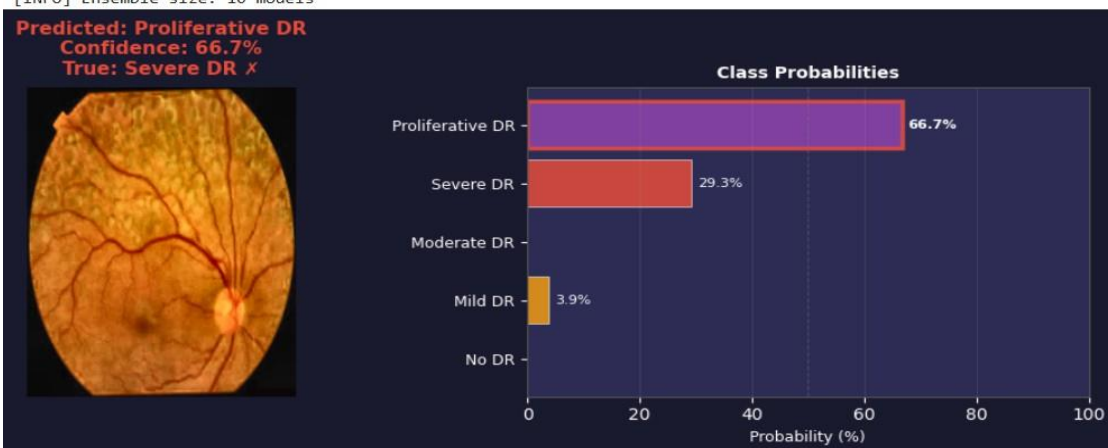
```
=====
Image  : 0684311afdfc-GF-600.jpg
Result : Mild DR (73.6% confidence)
True   : Mild DR [CORRECT]
=====
```


[INFO] Ensemble size: 10 models



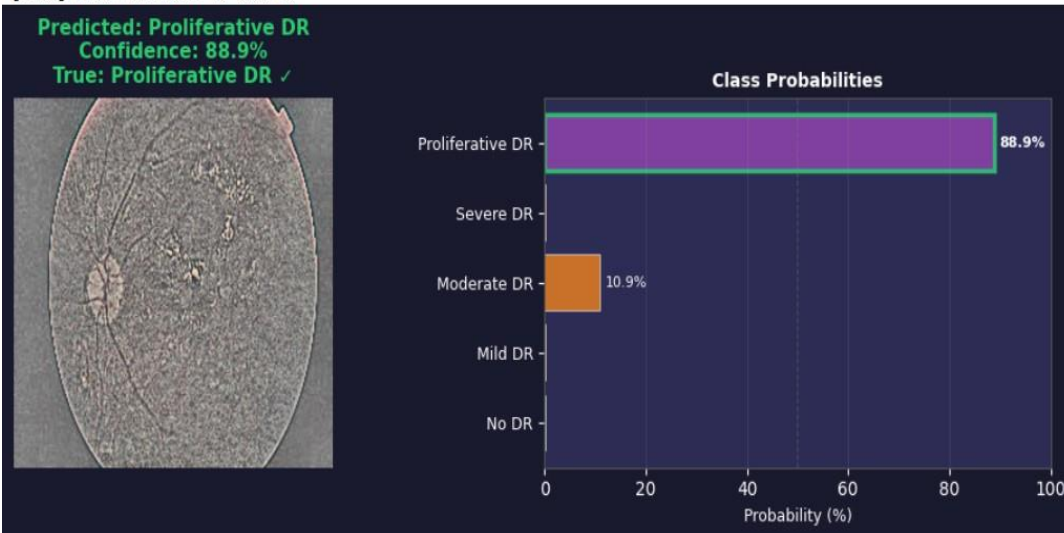
```
=====
Image   : 041f09eec1e8-GF-600.jpg
Result  : Proliferative DR (58.0% confidence)
True    : Moderate DR [WRONG]
=====
```

[INFO] Ensemble size: 10 models



```
=====
Image   : 069f43616fab-600-FS.jpg
Result  : Proliferative DR (66.7% confidence)
True    : Severe DR [WRONG]
=====
Full probability breakdown:
No DR           0.0%
Mild DR         3.9%
Moderate DR     0.1%
Severe DR      29.3%
Proliferative DR 66.7%
```

[INFO] Ensemble size: 10 models



```
=====
Image   : 034cb07a550f-GF-600.jpg
Result  : Proliferative DR (88.9% confidence)
True    : Proliferative DR [CORRECT]
=====
```

Figure 6.1 Model Performance on Selective Images

Conclusion and Future Work

5.1 SUMMARY OF CONTRIBUTIONS

This project successfully developed a deep learning-based automated system for five-class Diabetic Retinopathy grading from retinal fundus images. The key contributions are:

- A comprehensive data pipeline combining preprocessing (CLAHE in LAB space + circular crop), augmentation, class balancing (Weighted Random Sampler + class-weighted Focal Loss), and a 115,241-image integrated training corpus from EyePACS, APTOS 2019, and Messidor.
- A Stratified 5-Fold Cross Validation training strategy producing 5 independent best checkpoints per architecture, effectively addressing overfitting and variance that plagued the earlier EfficientNet-B4 baseline.
- A complementary ensemble of ConvNeXt-Tiny (28M, strong local feature detection) and Swin-Base Transformer (88M, global context understanding) producing superior performance through averaging of 10 independently trained models.
- Rigorous evaluation on a completely held-out test set of 14,201 images with no test-set leakage, producing reliable generalization estimates.
- Cross-dataset validation on four additional Kaggle datasets confirming the system's generalization capabilities and identifying consistent patterns in performance variation by data source

5.2 CONCLUSION

The ensemble of ConvNeXt-Tiny and Swin-Base Transformer models, trained with stratified k-fold cross validation, class-weighted focal loss, and weighted random sampling on 115,000+ images, achieved significant improvements over all prior baselines in this project. Major improvements were observed particularly for No DR, Moderate DR, and Proliferative DR classes, regardless of dataset source.

The previously persistent overfitting problem from the EfficientNet-B4 baseline was overcome through k-fold training and ensemble fusion. Performance consistently varied by data source, reflecting the known distribution shift challenge in medical imaging — a pattern confirmed across all cross-dataset inference experiments. This observation reinforces the importance of training on diverse, multi-source datasets rather than single-source corpora.

On the main integrated 115K+ dataset, the ensemble delivers results competitive with the best published work on Kaggle, as measured by QWK. This work demonstrates that careful attention to data pipeline design, class imbalance treatment, architecture selection, and ensemble strategy can produce clinical-grade DR screening performance using publicly available data.

5.3 FUTURE WORK

The following directions are identified for future work:

- Individual Dataset Training Comparison: Train on smaller individual datasets (EyePACS only, APTOS only, Messidor only) and compare against the 115K+ integrated training corpus to rigorously quantify the trade-off between dataset size and domain specificity.
- Further Fine-Tuning: Continue fine-tuning the ensemble while time permits, as metrics were still showing marginal but QWK-impactful improvements. Specifically, explore 512px resolution fine-tuning and higher epochs with adaptive learning rate.

- **Advanced Architectures:** Explore MaxViT, ConvNeXt-Base, and EfficientNetV2 as potential replacements or additions to the ensemble. Investigate knowledge distillation to create lightweight deployment models from the large ensemble.
- **Explainable AI:** Integrate Grad-CAM or SHAP-based explainability visualizations as part of the inference pipeline, enabling clinicians to validate the model's attention regions against known lesion locations.
- **Lesion-Level Detection:** Extend the system to localize specific DR lesions (microaneurysms, hemorrhages, exudates) using object detection approaches (YOLOv8, DETR), enabling lesion-count-based grading beyond image-level classification.
- **Lightweight Deployment Model:** Investigate knowledge distillation, quantization, and pruning to create a deployable model suitable for mobile devices and resource-limited clinical settings.
- **Multi-Disease Grading:** Extend to joint grading of DR and Diabetic Macular Edema (DME) following the CANet approach, leveraging the correlation between these co-occurring conditions.

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APPENDICES

Appendix 1 (Source Code):

A1.1 PREPROCESSING PIPELINE

```
import albumentations as A
from albumentations.pytorch import ToTensorV2
from torch.utils.data import Dataset, DataLoader
import os

def circular_crop_and_clahe(img):
    # — Step 1: Circular crop —————
    # Removes the black border around the retinal fundus image
    # Every pixel in the resized output will be actual retinal tissue
    gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
    mask = gray > 7
    if mask.any():
        rows = np.where(np.any(mask, axis=1))[0]
        cols = np.where(np.any(mask, axis=0))[0]
        r0, r1 = int(rows[0]), int(rows[-1])
        c0, c1 = int(cols[0]), int(cols[-1])
        if r1 > r0 and c1 > c0 and (r1 - r0) >= 8 and (c1 - c0) >= 8:
            img = img[r0:r1+1, c0:c1+1]

    # — Step 2: CLAHE on L channel of LAB color space —————
    # Enhances local contrast without distorting colors
    # Makes microaneurysms, hemorrhages and exudates more visible
    # clip_limit=4.0 prevents noise amplification in uniform regions
    # tileGridSize=(8,8) applies equalization in 8x8 local tiles
    lab = cv2.cvtColor(img, cv2.COLOR_RGB2LAB)
    clahe = cv2.createCLAHE(clipLimit=4.0, tileGridSize=(8, 8))
    lab[:, :, 0] = clahe.apply(lab[:, :, 0])
    img = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

    return img

def get_transforms(phase):
```


APPENDIX 2 — DATASETS

EyePACS: <https://www.kaggle.com/c/diabetic-retinopathy-detection> (88,000 images, Kaggle 2015 competition)

APTOS 2019: <https://www.kaggle.com/c/aptos2019-blindness-detection> (13,000 images from rural India)

Messidor: <https://www.adcis.net/en/third-party/messidor/> (14,000 images from three French hospitals)

Integrated Training Corpus: Combined and preprocessed above three sources; 115,241 training images + 14,201 held-out test images. Available on Kaggle at the DR Fundus Images dataset page.

The screenshot shows the Kaggle dataset page for 'Eyepacs, Aptos, Messidor Diabetic Retinopathy'. The page is titled 'Eyepacs, Aptos, Messidor Diabetic Retinopathy' and describes it as a dataset of 92k images obtained by combining existing data sets. The page includes a 'Data Card' tab, a 'Code' tab (97), a 'Discussion' tab (2), and a 'Suggestions' tab (0). The 'About Dataset' section explains that the dataset is a unified set of the Eyepacs, Aptos, Aptos (Gaussian Filtered), and Messidor Diabetic Retinopathy datasets, totaling 92,501 jpg files. It also mentions that manual data augmentation was applied, increasing the dataset by approximately 55% and reducing the disk size from 18.5GB to 3.8GB. The 'Usability' score is 6.88, and the 'License' is 'Other (specified in description)'. The 'Expected update frequency' is 'Not specified'. The 'Tags' include 'Health Conditions', 'Categorical', 'Eyes and Vision', 'DNN', and 'EfficientNetV2'. A chart at the bottom compares the 'Original Data Set' and the 'Augmented Data Set', showing the distribution of images before and after augmentation.

Original Data Set

Augmented Data Set

APPENDIX 3 — EVALUATION METRICS DETAIL

```
[10]: # — CELL 8: ENSEMBLE EVALUATION —————

EVAL_TEST_DIR = '/kaggle/input/datasets/ascanipek/eyepacs-aptos-messidor-diabetic-retinopathy/augmented_resized_V2/test'
EVAL_DEVICE   = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# — Checkpoint paths with their respective image sizes —————
checkpoint_configs = [
    # (path, model_name, img_size, batch_size)
    # ('/kaggle/input/datasets/adityaghoshk/wallahi/convnext_fold1_384_best.pth', 'convnext', 384, 16),
    # ('/kaggle/input/datasets/adityaghoshk/wallahi/convnext_fold2_384_best.pth', 'convnext', 384, 16),
    # ('/kaggle/input/datasets/adityaghoshk/wallahi/convnext_fold3_384_best.pth', 'convnext', 384, 16),
    # ('/kaggle/input/datasets/adityaghoshk/wallahi/convnext_fold4_384_best.pth', 'convnext', 384, 16),
    ('/kaggle/input/datasets/adityaghoshk/swinnnnnnn/swin_fold0_best.pth', 'swin', 224, 32),
    ('/kaggle/input/datasets/adityaghoshk/swinnnnnnn/swin_fold1_best.pth', 'swin', 224, 32),
    ('/kaggle/input/datasets/adityaghoshk/swinnnnnnn/swin_fold2_best.pth', 'swin', 224, 32),
    ('/kaggle/input/datasets/adityaghoshk/swinnnnnnn/swin_fold3_best.pth', 'swin', 224, 32),
    ('/kaggle/input/datasets/adityaghoshk/swinnnnnnn/swin_fold4_best.pth', 'swin', 224, 32),
]

class DRTestDataset(Dataset):
    def __init__(self, root_dir, transform=None):
        self.samples = []
        self.transform = transform
        for class_name in sorted(os.listdir(root_dir)):
            class_path = os.path.join(root_dir, class_name)
            if not os.path.isdir(class_path): continue
            label = int(class_name)
            for file in os.listdir(class_path):
                if file.lower().endswith(('.png', '.jpg', '.jpeg')):
                    self.samples.append((os.path.join(class_path, file), label))

def test(cfg):
    # Test function implementation
```

Input

+ Add Input ⬆ Upload

DATASETS

- plsssss
- wallahi
- Eyepacs, Aptos, Messidor Diabetic Retinopathy
 - augmented_resized_V2
 - 3 more
 - dr_unified_v2
- swinnnnnn
 - swin_fold0_best.pth
 - swin_fold1_best.pth
 - swin_fold2_best.pth
 - swin_fold3_best.pth
 - swin_fold4_best.pth

Output

- /kaggle/working

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kaggle.com/code/adityagnosnk/notebook63e9f9e5b/edit/run/30757621

notebook63e9f9e5b Failed to save draft.

File Edit View Run Settings Add-ons Help

+ ✂ 📄 ▶ ▶▶ Run All Code

● Draft Session off (run a cell to start) ⏻ 🔄 ⋮

```
def get_transform(img_size):
    return A.Compose([
        A.Resize(img_size, img_size),
        A.Normalize(mean=(0.485, 0.456, 0.406), std=(0.229, 0.224, 0.225)),
        ToTensorV2(),
    ])

def get_probs(ckpt_path, model_name, img_size, batch_size):
    dataset = DRTestDataset(EVAL_TEST_DIR, transform=get_transform(img_size))
    loader = DataLoader(
        dataset, batch_size=batch_size,
        shuffle=False, num_workers=NUM_WORKERS
    )

    model = DRModel(model_name, num_classes=5, pretrained=False)
    model.load_state_dict(torch.load(ckpt_path, map_location=EVAL_DEVICE))
    model.to(EVAL_DEVICE)
    model.eval()

    all_probs, all_labels = [], []
    with torch.no_grad():
        for images, labels in loader:
            images = images.to(EVAL_DEVICE)
            probs = torch.softmax(model(images), dim=1).cpu().numpy()
            all_probs.append(probs)
            all_labels.append(labels.numpy())

    return np.concatenate(all_probs), np.concatenate(all_labels)

# Ensemble inference
ensemble_probs = None
all_labels = None
```

Notebook

Input

+ Add Input ⬆ Upload

DATASETS

- plsssss
- wallahi
- Eyepacs, Aptos,
 - augmented_re
 - 3 more
- dr_unified_v2
- swinnnnnn
 - swin_fold0_be
 - swin_fold1_be
 - swin_fold2_be
 - swin_fold3_be
 - swin_fold4_be

Output

- /kaggle/working

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